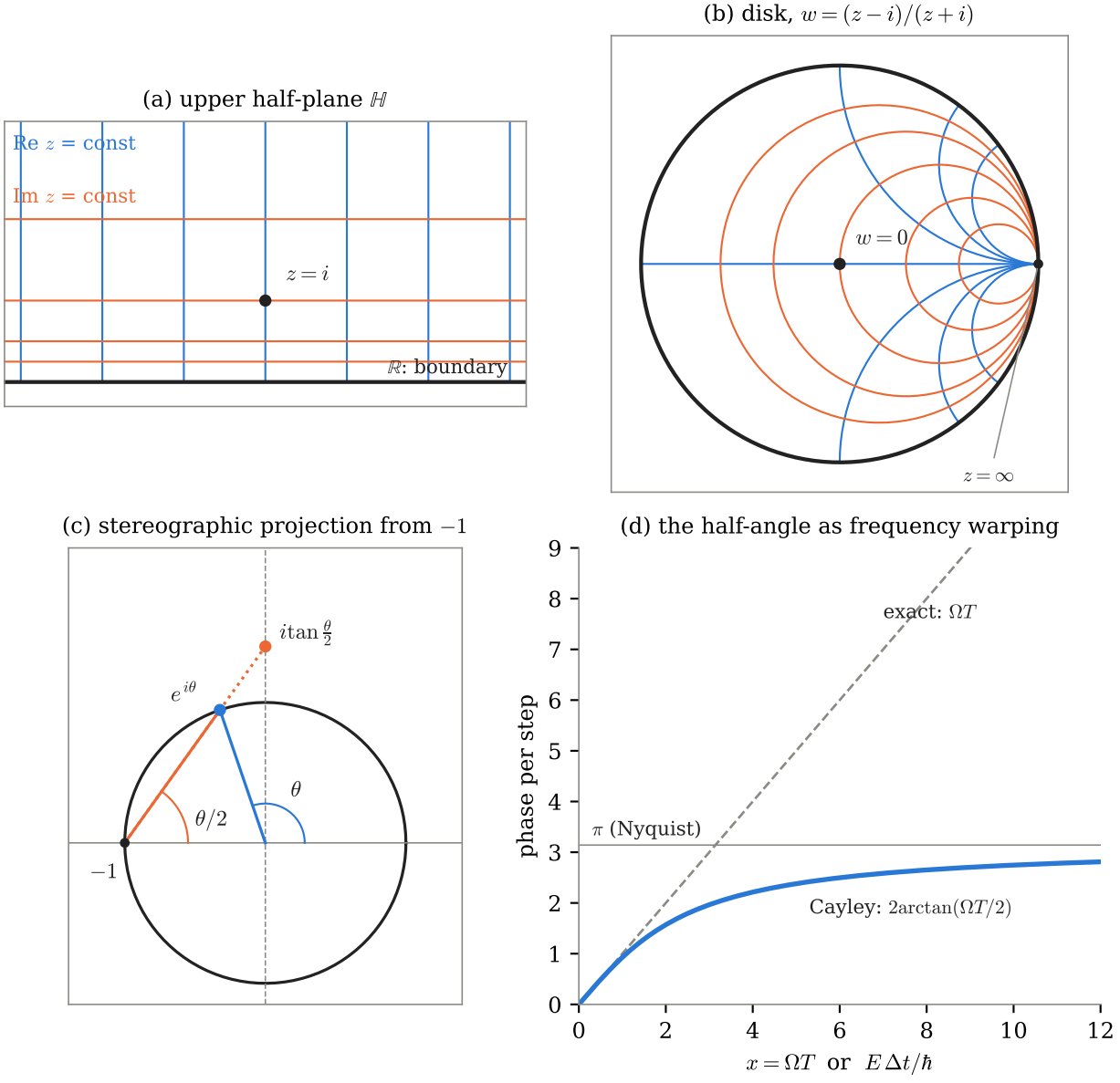
(a) upper half-plane $\mathbb{H}$
Re z = const
Im z = const
$z = i$
$\mathbb{R}$: boundary
(b) disk, $w = (z - i)/(z + i)$
$w = 0$
$z = \infty$
(c) stereographic projection from -1
$i\tan\frac{\theta}{2}$
$e^{i\theta}$
θ
$\theta/2$
-1
(d) the half-angle as frequency warping
exact: ΩT
π (Nyquist)
Cayley: $2\arctan(\Omega T/2)$
phase per step
$x = \Omega T$ or $E\,\Delta t/\hbar$
0
1
2
3
4
5
6
7
8
9
0
2
4
6
8
10
12